AI-Powered Student Performance Prediction & Early Warning System

Abstract

This project uses Machine Learning to identify students who are academically at risk before final examinations. The system analyzes academic and behavioral data such as attendance, quiz scores, assignment performance, internal examination marks, previous academic records, participation, homework completion, study hours, and parent meeting information.

Based on these features, the system predicts whether a student belongs to the Low Risk, Medium Risk, or High Risk category. It also provides personalized recommendations to help teachers intervene early and improve student performance.

Problem Statement

Teachers usually identify weak students only after final examinations. By that time, it becomes difficult to improve their academic performance.

This project aims to predict student performance in advance so that teachers can take timely corrective actions.

Objectives

- Predict student academic risk using Machine Learning.
- Identify weak students before final examinations.
- Generate early warning alerts.
- Provide personalized recommendations for improvement.
- Support teachers in making data-driven decisions.

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

- Streamlit

Machine Learning Algorithm

Random Forest Classifier

Reason:

- High accuracy
- Handles multiple features effectively
- Reduces overfitting
- Easy to interpret using feature importance

Dataset Features

- Attendance
- Quiz Score
- Assignment Score
- Internal Exam
- Previous Percentage
- Participation
- Homework Completion
- Behaviour Score
- Study Hours
- Parent Meeting
- Risk Level (Target)

Project Workflow

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Student Dataset

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Data Cleaning

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Feature Selection

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Train-Test Split

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Random Forest Classifier

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Risk Prediction

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Early Warning System

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Recommendations
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Results

The model successfully classifies students into:

- Low Risk
- Medium Risk
- High Risk

Teachers receive early warning alerts along with personalized recommendations to improve student performance.

Future Scope

- School ERP Integration
- Parent SMS and Email Alerts
- Cloud Deployment
- Mobile Application
- Multi-School Dashboard
- Deep Learning Models

Conclusion

This project demonstrates how Artificial Intelligence can assist teachers in identifying academically at-risk students before final examinations. By providing early predictions and personalized recommendations, the system supports timely intervention and helps improve overall student performance.